

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)			(warm with) aqueous sodium hydroxide (1) add (dilute nitric) acid (to neutralise the solution) (1) add aqueous silver nitrate (1) award (1) for all three precipitate colours white precipitate shows that the halogen is chlorine cream precipitate shows that the halogen is bromine yellow precipitate shows that the halogen is iodine accept correct observations for solubility of precipitates in ammonia solution (as an alternative to the mark for colours) precipitate soluble in aqueous ammonia shows that the halogen is chlorine precipitate partially soluble in aqueous ammonia / soluble in concentrated ammonia solution shows that the halogen is bromine precipitate insoluble in aqueous/concentrated ammonia shows that the halogen is iodine	4			4		4

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			Mass spectrum molecular ion peak(s) $\Rightarrow M_r = 92/94$ (1) ratio of peak heights 3:1 for molecular ion peaks $\Rightarrow$ Cl present (1) [accept also for atomic peaks at 35/37] award (1) for any correct fragment on mass spectrum e.g. peak at 15 $\Rightarrow$ CH₃ peak at 29 $\Rightarrow$ C₂H₅ ¹³C NMR spectrum 4 peaks $\Rightarrow$ 4 carbon environments (1) molecular formula $\Rightarrow$ C₄H₉Cl (1) ¹H NMR spectrum 4 peaks $\Rightarrow$ 4 hydrogen environments (1) award (1) for any one peak identified peak at 4.0 $\Rightarrow$ HC—Cl peak at 1.5 $\Rightarrow$ —CH₃ peak at 1.3 $\Rightarrow$ R—CH₂—R peak at 0.9 $\Rightarrow$ R—CH₃ award (1) for number of hydrogens in each environment linked to peak heights award maximum of 7 out of these 8 marks final mark reserved for correct structure of X $\Rightarrow$ 2-chlorobutane (1)	1					
					1					
						3	3	8		
				Question 11 total	6	3	3	12	0	4